Chapter 30 Problem Set

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1. Let's say we have $y \in \mathbb{R}$. We need to show that $\lim_{x \to y} x^2 = y^2$. By the limit laws, we know that equals $(\lim_{x \to y} x)^2$. We claim that $\lim_{x \to y} x = y$. If $\delta = \epsilon$, then $|y - x| < \delta$ implies $|y - x| < \epsilon$. So $\lim_{x \to y} x^2 = y^2$.

2. Let's say we have $y \in \mathbb{R}$. Let $\delta = \frac{\epsilon}{2}$. Then $|(y+1) - (x+1)| = |y - x| < \frac{\epsilon}{2} < \epsilon$ and $|(2y+1) - (2x+1)| = 2|y - x| < 2\frac{\epsilon}{2} = \epsilon$. In both cases, $|f(y) - f(x)| < \epsilon$.

3. Let's say we have $y \in \mathbb{R}$. Since both $f(x)$ and $g(x)$ are continuous, there exists δ_1 such that for all $\epsilon > 0$ and $x < 0$, if $|y - x| < \delta_1$ then $|f(y) - f(x)| < \epsilon$, and there exists δ_2 such that, for $x \geq 0$, if $|y - x| < \delta_2$ then $|g(y) - g(x)| < \epsilon$. Let $\delta = \min(\delta_1, \delta_2)$. Given that $|y - x| < \delta$, we can conclude that $|h(y) - h(x)| < \epsilon$ for any $\epsilon > 0$ and $x \in \mathbb{R} \backslash \{0\}$.

Now, it remains to show that $\lim_{x \to 0} h(x) = h(0) = 0$. If we are given that $|x| < \delta = \min(\delta_1, \delta_2)$, then when $x < 0$, $|h(x)| = |f(x)| < \delta$, and when $x \geq 0$, $|h(x)| = |g(x)| < \delta$. In either case, $|h(x)| < \delta$.

4. The limit does not exist. Let $\epsilon = \frac{1}{2}$. We see that the function is 1 for $x > 3$ and -1 for $x < 3$. So if we are given that $|x - 3| < \delta$, there exists no s such that $|\frac{|x-3|}{x-3} - s| < \frac{1}{2}$ because if s is within $\frac{1}{2}$ of 1 then it is not within $\frac{1}{2}$ of -1, which means that $|\frac{|x-3|}{x-3} - s| < \frac{1}{2}$ isn't guaranteed.

5. We need to show that there exists N such that, if $n, m > N$, then $|f(a_n) - f(a_m)| < \epsilon$ for all $\epsilon > 0$. We are given that $|a_n - a_m| < \epsilon$ for all $\epsilon > 0$. Also, since f is continuous, there exists $\delta > 0$ such that if $|x - y| < \delta$ then $|f(x) - f(y)| < \epsilon$. Combining both facts, we see that there exists $\delta > 0$ such that $|a_n - a_m| < \delta$, which implies that $|f(a_n) - f(a_m)| < \epsilon$. So that statement is guaranteed if $N < \min(m, n)$.

6. Using the limit laws, we see that $\lim_{x \to 0} xf(x) = (\lim_{x \to 0} x)(\lim_{x \to 0} f(x)) \leq B \lim_{x \to 0} x = 0$. Since $|\sin\left(\frac{1}{x}\right)| \leq 1$, we get that $\lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0$.

7. The function f is continuous only at $x = 0$. Consider the limit as x approaches a. Both rational and irrational values exist that are arbitrarily close to a, so for the limit to exist, the limit when considering only rationals and only irrationals must agree. Looking at only rational values, $f(x) = x$ so $\lim_{x \to a} f(x) = a$. Looking at irrational values, $f(x) = 0$ so $\lim_{x \to a} f(x) = 0$. These two limits only the same when $a = 0$, so the function is only continuous at $x = 0$.

8. First, let's assume f is continuous at a. This means that for every $\epsilon > 0$, there exists $\delta > 0$ such that if $|x - a| < \delta$ then $|f(x) - f(a)| < \epsilon$. Now, consider any sequence $x_1, x_2, \ldots$ with $\lim_{n \to \infty} x_n = a$. This means that for every $\delta > 0$, there exists N such that if $n > N$ then $|x_n - a| < \delta$. Combining these facts, we get that for $n > N$, we have $|x_n - a| < \delta$ which implies that $|f(x_n) - f(a)| < \epsilon$. So if we let $b = f(a)$, we have $\lim_{n \to \infty} f(x_n) = b$.

Now, let's assume that for every sequence $x_1, x_2, \ldots$ with $\lim_{n \to \infty} x_n = a$ we have $\lim_{n \to \infty} f(x_n) = b$. This means that if $n > N$ implies $|x_n - a| < \delta$ then it also implies that $|f(x_n) - f(a)| < \epsilon$. Therefore, $|x_n - a| < \delta$ implies that $|f(x_n) - f(a)| < \epsilon$. Since this is true for all sequences in $\mathbb{R}$ converging to a, it is true for any $x \in \mathbb{R}$. That means that if $|x - a| < \delta$ then $|f(x) - f(a)| < \epsilon$, so f is continuous at a.